Mohana Dhara

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SUMMARY

Performance-Driven Software Engineer with 4+ years of experience in the full SDLC, engineering high-availability, cloud-native Microservices and Distributed Systems using Java (Spring Boot) and Apache Kafka.

Expert in building End-to-End Full-Stack applications (React/Angular) and Event-Driven Architectures on AWS, Azure, and GCP to support Scalable transaction processing for 21k+ daily events.

Proven track record in an Agile environment of optimizing System Design and reducing Time-to-Market by 34% through CI/CD automation, Kubernetes orchestration, and Performance Tuning

SKILLS

Languages: Java, JavaScript, TypeScript, Python, Kotlin, SQL, Bash

Backend & Frameworks: Spring Boot, Spring Cloud, Spring Security, Hibernate, JPA, Node.js, Express.js, GraphQL, RESTful APIs, Reactive Programming

Frontend: React.js, Angular, Next.js, Redux Toolkit, HTML5, CSS3, Tailwind CSS, Webpack, Vite, SSR, PWA

Cloud Platforms: AWS, Microsoft Azure, GCP

DevOps & Infrastructure: Docker, Kubernetes, Jenkins, GitHub Actions, GitLab CI/CD, Nginx, Infrastructure as Code (IaC), Linux, GitOps

Databases & Messaging: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Apache Kafka, Kafka Streams, RabbitMQ

Architecture & Design: Microservices, Event-Driven Architecture, Distributed Systems, System Design, Design Patterns, Caching Strategies

Testing & Security: JUnit, Mockito, Testcontainers, Cypress, Selenium, OAuth 2.0, OpenID Connect, API Security, Secure Coding

Tools & Practices: Git, JIRA, Agile, Performance Optimization

PROFESSIONAL EXPERIENCE

Software Engineer, *Wells Fargo* 08/2024 – Present Remote, USA

- Engineering Spring Boot microservices for a real-time event monitoring platform, designing RESTful APIs and MongoDB data models to support high-throughput data evaluation across 9 cross-functional engineering teams.
- Developing Angular dashboard using RxJS and Reactive Forms to support fraud investigation workflows, enabling teams to review event histories and case data across 3 distributed processing systems.
- Implementing CI/CD pipelines with Jenkins automating build validation, testing, and Docker image publishing for Spring Boot services, reducing deployment cycle duration by 34% across fraud-monitoring platform environments.
- Containerized Spring Boot applications with Docker and deployed services on AWS infrastructure, supporting scalable event-processing pipelines handling 21k payment authorization events daily across distributed banking channels.
- Built Kafka streaming pipelines capturing high-volume application events, enabling near real-time monitoring workflows and reducing data processing lag by 26% across distributed microservices.
- Designed MongoDB indexing and document modeling strategies for high-volume transaction records, improving investigation query performance for fraud-analysis APIs supporting distributed risk-monitoring services.
- Integrated secure microservice communication using OAuth2, JWT authentication, and AWS IAM roles through controlled API access policies, strengthening protected transaction data exchange across fraud-detection platform services.

Software Engineer, *HCL Tech* 04/2022 – 12/2023 Mumbai, India

- Architected and developed Spring Boot microservices for an enterprise digital operations platform, designing RESTful APIs and PostgreSQL data models enabling scalable workflow automation across distributed business applications.
- Built Angular-based analytics dashboard using JavaScript to visualize operational metrics, service workflows, and platform usage insights, enabling internal teams to monitor enterprise application performance and process efficiency.
- Engineered CI/CD pipelines using Azure DevOps to automate build validation, automated testing, and container artifact publishing for Java microservices, reducing deployment turnaround time by approximately 32%.
- Containerized Spring Boot applications using Docker and orchestrated deployments through Kubernetes on Microsoft Azure, enabling highly available service-processing pipelines supporting enterprise-scale workflow transactions.
- Implemented Apache Kafka event-driven architecture enabling asynchronous communication across microservices, improving system responsiveness and reducing cross-service processing latency across distributed enterprise applications.
- Optimized PostgreSQL database performance through advanced indexing, partitioning, and query tuning strategies, significantly improving reporting and data retrieval performance across large-scale operational datasets.
- Strengthened platform security by integrating token-based authentication and API gateway policies for secure inter-service communication, ensuring controlled access to enterprise application services and sensitive operational data.

Software Development Engineer - Research, *SRM University* 01/2021 – 03/2022 AP, India

- Designed Spring Boot microservices for a campus research collaboration portal, implementing REST APIs and MySQL schema supporting researcher profiles, projects, and publications across 6 academic departments.
- Built React dashboard using JavaScript and Redux Toolkit to visualize experiment datasets and collaboration metrics, improving internal data accessibility for 40+ research students managing shared datasets.
- Implemented CI/CD pipelines using Jenkins and GitHub Actions, automating build, testing, and container deployment workflows, reducing manual release effort by 35% across development and staging environments.
- Containerized Spring Boot services with Docker and deployed on Kubernetes within Google Cloud Platform, supporting scalable experiment-data pipelines handling 15k+ dataset requests monthly.
- Developed asynchronous Kafka workflows to process research dataset updates and notification events, reducing cross-service data synchronization delays by about 28% across distributed microservices.
- Optimized MySQL queries and integrated Redis caching for dataset retrieval APIs, improving response latency for large experiment metadata queries across microservice-based research data platforms.

EDUCATION

Master of Science, *University of North Carolina* *May 2025 Charlotte*, USA
Computer Science, GPA: 3.6

Bachelor of Engineering, *SRM University* *Aug 2023 Amaravati*, India
Computer Science and Technology

PROJECTS

Task Management System
Java, Spring Boot, Spring Security, JWT, REST API, JPA/Hibernate, MySQL

- Engineered a full-stack RESTful task management application using Spring Boot, enabling users to create, update, prioritize, and track tasks with role-based access control
- Implemented JWT-based authentication and authorization using Spring Security, securing all API endpoints and managing stateless user sessions
- Designed and exposed aggregate REST endpoints to provide summarized task views (e.g., tasks by status, priority, and deadline) using JPQL and Spring Data JPA
- Built a normalized relational schema in MySQL with entity relationships (Users ↔ Tasks ↔ Categories), leveraging Hibernate ORM for seamless data persistence
- Applied DTO pattern and input validation (Bean Validation / @Valid) to enforce data integrity and decouple API layer from persistence layer

Anti-Fraud System
Java, Spring Boot, Spring Security, REST API, Role-Based Access Control, Heuristic Algorithms

- Developed a transaction fraud detection system using Spring Boot with a multi-tier role model (Administrator, Merchant, Support), enforcing fine-grained access control via Spring Security
- Implemented heuristic-based transaction validation rules (IP blacklisting, stolen card detection, transaction amount thresholds) to automatically flag, block, or allow transactions in real time
- Designed a suite of secured REST endpoints for user management, transaction submission, and fraud reporting, following RESTful conventions with proper HTTP status codes
- Built an adaptive threshold mechanism that adjusts allowed transaction limits based on transaction feedback history, simulating real-world fraud model tuning
- Managed complex entity relationships across Users, Transactions, Suspicious IPs, and Stolen Cards using Spring Data JPA with a relational database.

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate
- Microsoft Azure Fundamentals (AZ-900)
- AI Programming with Python Nano degree (AWS x Udacity)